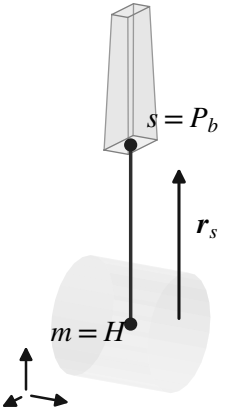
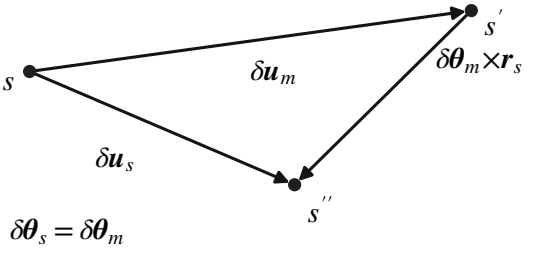
(a)


$s = P_b$
$\boldsymbol{r}_s$
$m = H$

(b)


s
s'
s''
$\delta \boldsymbol{u}_m$
$\delta \boldsymbol{\theta}_m \times \boldsymbol{r}_s$
$\delta \boldsymbol{u}_s$
$\delta \boldsymbol{\theta}_s = \delta \boldsymbol{\theta}_m$